DATA ITEM DESCRIPTION
Title: ELECTRICAL POWER CAPACITY REPORT

Number: DI-SESS-82288
AMSC Number: N10072
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20190815
Limitation: None
GIDEP Applicable: N/A
Project Number: SESS-2019-038

Use/Relationship: The Electrical Power Capacity Report documents the capability of the electrical power system to produce and distribute the power required by the utilization equipment.
Standard Electric Load Analysis charts are provided in accordance with MIL-E-7016 for maintaining electrical load configuration for existing, modified, and new electrical power systems. The data will be used to verify the interoperability of electrical power sources and utilization equipment.

This Data Item Description (DID) contains the format, content, and intended use information for the Electrical Power Capacity Report resulting from the work task described by the contract statement of work.

Requirements:

1. Reference Documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions shall be as specified in the solicitation or contract.

(Copies of the Military Specification, MIL-E-7016, Analysis of Aircraft Electric Load and Power Source Capacity are available online at https://quicksearch.dla.mil/)

2. Format. Contractor's format is acceptable.

3. Content. The report shall include the following information:

3.1 Electric loads and power source analysis charts for each power source utilized by the system equipment prepared in accordance with MIL-E-7016, with the additional information:

3.1.1 Connected loads chart

3.1.1.1 The connected loads chart (Section 3.5.4.1) shall include columns for the Load Type (e.g. Linear, Motor, Constant Power) and the Identifier for the connector or harness that supplies power to the utilization equipment.

3.1.1.2 Any symbol may be used in column D (Section 3.5.4.1.4.1) to denote leading power factor, as long as the symbol is clearly identifiable. The analysis shall document the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:01Z
Check the source to verify that this is the current version before use.

meaning of the selected symbol and consistently use the same symbol to denote leading power factor across the entire analysis.

3.1.1.3 Loads that operate for longer than the time interval being analyzed and listed in column F (Section 3.5.4.1.6) shall be considered continuous. The analysis shall document the duration of continuous loads as the duration of that time interval.

3.1.2 The loads analysis chart shall include:

3.1.2.1 Power utilization information for all equipment drawings power from system power buses

3.1.2.2 Operating conditions of the system, including each of the system States and Modes as well as power source failure conditions

3.1.2.3 Connected load charts, including equipment duty cycle for each load in each state

3.1.2.4 Transient load profiles associated with distinct operating conditions for each item listed as multiple entries for each item

3.1.2.5 Discharge profiles of bus hold up batteries within the system, including the time to reach the minimum operating voltage for the connected loads in each operating condition

3.2 A generation capacity analysis for system generated power during operating conditions, including source ratings for:

3.2.1 Steady state conditions

3.2.2 Normal transient conditions

3.2.3 Overload conditions

3.2.4 Fault conditions

3.3 A distribution system analysis, including the capacity and power losses of the distribution system during the following conditions:

3.3.1 Nominal loads included in the electric loads analysis

3.3.2 Overloads

3.3.3 Fault Conditions

End of DI-SESS-82288

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:01Z
Check the source to verify that this is the current version before use.